Salomón Hernández Velandia

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portfolio-salomonhdz-dev.vercel.app

Languages

Español — C2

English — B1

Publications

Comparative study of methods for generating echocardiographic images
IEEE — SIPAIM 2025
December 18th, 2025

Comparative evaluation of StyleGAN2-ADA, MedGAN, VQ-GAN, and Pix2Pix for synthetic echocardiographic image generation under comparable training conditions. The study evaluated image quality and model performance using quantitative and qualitative metrics, identifying differences in convergence, structural fidelity, and generation quality.

Keywords: Generative AI · GANs · Deep Learning · Computer Vision · Medical Imaging

DOI: 10.1109/SIPAIM67325.2025.11283212

Profile

Python Developer with 1+ year of professional experience developing software solutions and automation for a U.S.-based company, complemented by research experience in Artificial Intelligence and Machine Learning. Experience working with Python, SQL, data processing, automation, REST APIs, and web development, with additional experience in data cleaning, analysis, visualization, and machine learning through research and technical projects. Experienced in remote and international work environments, with an IEEE publication in Artificial Intelligence and generative models.

Professional Experience

Universidad Nacional de Colombia

Generative Models Medical Researcher

July 2025 – February 2026 | Bogotá, Colombia

Research project focused on applying deep learning and generative models to synthetic medical image generation, with the goal of increasing the availability of echocardiographic datasets for AI-based medical research.

- Developed and evaluated generative AI workflows for creating synthetic echocardiographic images using generative and image reconstruction models to augment medical datasets.
- Designed and implemented data processing, model training, and evaluation workflows using Python, TensorFlow, PyTorch, NumPy, and Pandas.
- Trained and evaluated generative models including StyleGAN2-ADA, MedGAN, Pix2Pix, and VQGAN using the EchoNet-Dynamic dataset from Stanford University.
- Tuned hyperparameters and compared different deep learning architectures and configurations to identify effective approaches for synthetic image generation.
- Evaluated model performance using FID, KID, SSIM, PSNR, LPIPS, and JSD, combining quantitative metrics with statistical analysis and data visualization.
- Documented and presented research findings through scientific publications and presentations at national and international conferences.
- Published research results in IEEE and presented findings at national and international conferences.

Tecnologías: Python, TensorFlow, PyTorch, NumPy, Pandas, OpenCV, Matplotlib, Deep Learning, Generative AI, GANs, Computer Vision, Medical Imaging, Data Analysis.

Bilingual Child Care Training

Software Developer

April 2024 – July 2025 | United States

Software Developer responsible for the design, development and maintenance of digital solutions for an educational technology platform, working on web applications, AI-based services, API integrations and Moodle-based learning systems.

Skills

Programming Languages
- Python
- PHP
- JavaScript
- C++

Machine Learning
- TensorFlow
- PyTorch
- Scikit-Learn
- Keras
- OpenCV
- GANs
- Computer Vision
- Deep Learning
- Pattern Recognition

Data Science
- Pandas
- NumPy
- Matplotlib
- Power BI
- ETL
- Data Cleaning
- Data Visualization
- Feature Engineering
- Statistical Analysis

Databases & Query Languages
- SQL
- MySQL
- MariaDB

Backend
- REST APIs
- Google APIs
- Moodle
- Strapi
- Make
- Docker
- Git

Frontend
- React
- HTML
- CSS
- SASS
- Mustache

Multimedia
- Photoshop
- Illustrator
- Figma
- UI/UX Design
- 3D Animation
- 3D Modeling

- I developed AI solutions using Python and OpenAI Whisper to automate transcription, translation, and multilingual educational content generation processes.
- I designed and integrated REST APIs from Google Meet, Gmail, and Google Drive to automate communication flows, data synchronization, and internal processes.
- I managed and optimized data flows using Strapi CMS, integrating multiple REST APIs and designing communication flows between backend services.
- I developed and customized solutions for Moodle, modifying themes, templates, and components using PHP, Mustache, SQL, HTML, CSS, and JavaScript.
- I led the development and customization of a Moodle-based educational platform supporting over 60 online courses, implementing functionalities and improvements in both the frontend and backend according to business requirements.
- I worked with MariaDB/MySQL and databases managed by Strapi, performing queries, data management, and integration with applications and services.
- I developed web applications and digital services using React, JavaScript, PHP, and SASS, including frontend interfaces and mobile solutions for end users.
- I worked remotely in an Agile/Scrum environment, using Git, Jira, and Zoho Projects for version control, task management, issue resolution, and production support.

Tecnologías: Python, OpenAI Whisper, React, JavaScript, PHP, Moodle, Strapi CMS, SQL, MariaDB, API REST, API de Google, Git, Jira, Zoho Projects, SASS, HTML, CSS, Agile/Scrum, Amazon S3.

Education

Universidad Militar Nueva Granada
Ingeniería Multimedia

January 2020 – 2026 | Bogotá, Colombia

Degree completion pending English proficiency requirement.

Academic focus: Signal and image processing, computer vision, pattern recognition, machine learning, deep learning, generative models, data analysis and visualization, with additional training in software development, databases, mathematics, statistics, UI/UX and product design.

Stanford Online
Machine Learning Specialization

July 2023 – April 2024

Stanford Online
Advanced Learning Algotithms: Neuronal Networks

February 2025 – April 2025

Certificates

Supervised Machine Learning: Regression and Classification

Advanced Learning Algorithms

SIPAIM 2025 — IEEE

Presented research on comparative methods for synthetic echocardiographic image generation using generative and reconstruction models.

Congreso Colombiano de Computación

Presented research findings related to generative models and synthetic medical image generation.

Projects

Crypto Trading Dashboard & Signal Engine

Developed a Python-based cryptocurrency market analysis platform combining historical and real-time market data with quantitative logic to generate buy and sell signals. Implemented data processing, analysis, visualization, and signal-generation workflows.

Technologies: Python, Pandas, NumPy, Matplotlib, Git, Data Analysis, Data Visualization.

Echocardiogram Image Generation with Generative AI

Thesis | Research Project

Developed a generative AI workflow for synthetic echocardiographic image generation, experimenting with multiple deep learning and generative architectures to augment medical imaging datasets. Implemented data processing, model training, experimentation, and evaluation workflows using Python and deep learning frameworks.

Technologies: Python, PyTorch, TensorFlow, Pandas, NumPy, Matplotlib, Git, Deep Learning, GANs, Computer Vision.

English Vocabulary Learning Application

Developed an interactive Python application for English vocabulary learning, including word input, automated text analysis, scoring, and immediate feedback through a graphical user interface.

Technologies: Python, GUI Development, Git.